MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS
FALL SEMESTER 2025
STUDY GUIDE FOR TEST 2

Chapter 3: Limits

(1) Definition of cluster points, limits, sequential criteria
(2) How to show uniqueness of limits
(3) Boundedness and limits
(4) Basic operations and Squeeze Theorem
(5) Definitions of one-sided limits and limit at infinity
(6) Some known limits: $\lim_{x\to 0} \frac{\sin(x)}{x} = 1$ and $\lim_{x\to 0} \frac{e^x-1}{x}$

Chatper 3: Continuity

(1) Definition of a continuous function. Relation to limit. Sequential criteria.
(2) Some basic examples of continuous functions, and some examples of discontinuous functions (e.g. example that left-hand limit is not equal to the right hand limit, example that left-hand limit equal to right hand limit, example that left-hand limit and right-hand limit does not exist)
(3) What kinds of properties do you have for a function on a closed bounded domain? Boundedness, maximum-minimum value theorem. The main idea in proving the max-min theorem.
(4) State the Bolzaono's intermediate value theorem, and the bisection method.
(5) Some practical ways to use intermediate value theorem for finding a root.
(6) Define the meaning of uniform continuity. Give examples of continuous functions which are not uniform continuous.
(7) State the uniform continuity theorem. How to prove the theorem.
(8) The definition of Lipschitz functions. Examples of Lipschitz functions. How to show Lipschitz functions are uniformly continuous.

Chapter 4: Differentiation

(1) Definition of derivative. How to find derivatives from the first principle or limit.
(2) How to show differentiability implies continuity.
(3) Examples of non-differentiable functions .
(4) Basic rules in computing derivatives. Chain Rule, and how Carathéodory's Theorem is used in showing Chain Rule. How to use chain rule to find derivatives of some invereses.
(5) State the Relative/Interior Extremum Value Theorem.
(6) Rolle's Theorem. State Mean-Value Theorem and what is the relation with Rolle's Theorem.

(7) How to use Mean-Value Theorem to relative first derivative and the monotone behavior of a function
(8) State Darboux's theorem. Connection to intermediate value theorem. Some examples and some applications.
(9) Use Relative/Interior Extremum Value Theorem, Mean-Value Theorem and Darboux's theorem for some practical problems e.g. showing existence of some c satisfying certain value of f or its derivatives